Sanya Sinha

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Education

Kalinga Institute of Industrial Technology — Bhubaneswar, Odisha
B.Tech in Computer Science Engineering (AI & ML) ***CGPA: 8.41*** — *2024 – 2028*

St. John's School — Jaunpur, Uttar Pradesh
Class XII ***85.2%*** — *2021 – 2023*

St. John's School — Jaunpur, Uttar Pradesh
Class X ***85%*** — *2009 – 2021*

Projects

Loan Approval Prediction System | *Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn*
- Built a classification system to predict loan Approved/Rejected status from applicant data, aiming to reduce manual verification bias and risky approvals.
- Performed data profiling, missing value imputation, EDA, categorical encoding, correlation heatmap analysis, and feature scaling.
- Trained and compared Logistic Regression, KNN, and Naive Bayes; selected Naive Bayes as the best model with **79% precision** to minimize false approvals.

SmartCart Customer Segmentation (Unsupervised ML) | *Python, Pandas, Scikit-learn, Matplotlib, Seaborn*
- Segmented a dataset of **2,236 customers with 19 features** using K-Means clustering on purchasing behaviour, engagement, and loyalty indicators.
- Handled missing values and outliers, engineered new features, and applied encoding and scaling before modeling.
- Determined optimal $\mathbf{k = 4}$ using the Elbow Method and Silhouette Score; profiled each cluster to identify high-value and churn-prone customer groups for targeted marketing.

House Price Prediction | *Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn*
- Engineered features and handled missing values in a housing dataset of **2,918 records** to prepare it for regression modeling.
- Built Linear, Ridge, and Lasso Regression models and evaluated them using cross-validation; best model achieved $\mathbf{R^2}$ **of 0.346** and **Adjusted** $\mathbf{R^2}$ **of 0.305**.

Titanic Survival Prediction | *Python, Pandas, Scikit-learn, Matplotlib, Seaborn*
- Applied train-test split and standardization, then compared Logistic Regression, KNN, Naive Bayes, Decision Tree, Random Forest, and Bagging classifiers.
- Reduced overfitting in Decision Trees using pre-pruning and post-pruning; evaluated models using accuracy, precision, recall, and F1-score.
- Best model: **Decision Tree with 84% accuracy**.

Technical Skills

Languages: Python, Java, C, SQL, HTML, CSS
Machine Learning: Classification (Logistic Regression, KNN, Naive Bayes, Decision Tree, Random Forest, Bagging), Regression (Linear, Ridge, Lasso), Clustering (K-Means), L1/L2 Regularization, Cross Validation, GridSearchCV, Feature Engineering, EDA
Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, BeautifulSoup
Tools: Git, GitHub, Jupyter Notebook, VS Code

Experience

Organising Committee Member — Nov. 2025 – Present
Institute of Electrical and Electronics Engineers (IEEE CTSoC), KIIT — *Bhubaneswar, Odisha*
- Organized technical events, workshops, and student engagement activities.
- Coordinated event logistics and supported the execution of an international research event.

Event Anchor — Dec. 2025 – Present
Kodessa (Women Tech Society of KIIT) — *Bhubaneswar, Odisha*
- Hosted technical sessions and managed stage communication during technical events.

General Volunteer – Slum Visit — Aug. 2024 – Present
The Philanthropy Hand (TPH), KIIT — *Bhubaneswar, Odisha*
- Participated in community outreach and social service initiatives for underprivileged communities.

Leadership & Campus Involvement

Kalakaar (Dramatic Society of KIIT) — Dec. 2024 – Present
Actor & Team Lead — *KIIT University*
- Led the team during rehearsals and national-level cultural competitions.
- Secured **Second Runner-Up** at **Spring Fest 2026, IIT Kharagpur** and **Runner-Up** at **NLU Odisha** in Nukkad Natak competitions.

Kinetix — Nov. 2025 – Present
Marketing Lead — *KIIT University*
- Led marketing campaigns and promotional activities for campus events.